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What Do Future Teachers Look For in Movement-Based Learning? Implications For Multimodal Systems

Allison Poh[0000−0002−2258−4532] and Ivon Arroyo[0000−0002−9697−8016]

University of Massachusetts Amherst, Amherst MA 01002, USA
{apoh,ivon}@cs.umass.edu

Abstract. Movement-based and embodied learning activities generate rich multimodal student data, but little is known about how teachers may use this complex information to interpret understanding and guide instruction. In this study, 25 preservice PreK–6 teachers participated in six movement-based math activities and reflected on: (1) interpreting student behavioral signals (body movements, facial expressions, hand gestures, and speech) as evidence of understanding or lack of understanding, (2) responding to concerning student behaviors, and (3) leveraging multimodal data to monitor and support students in real time. Reflections were collected through post-activity questionnaires and analyzed using inductive coding supported by affinity diagramming and descriptive statistics. The results suggest that future teachers interpret student understanding and lack of understanding by attending to signals across eight behavioral dimensions: affect, correctness, focus and engagement, intention, peer interaction and collaboration, rhythm and confidence, support use, and technology use. In addition, findings suggest that future teachers are generally comfortable with multimodal student data collection and perceive sources capturing body movement and speech as especially useful for monitoring and supporting students, particularly for their understanding of math concepts, the activity, and the movement–math connection. Overall, these findings illustrate how future teachers may use multimodal data in their classrooms and provide guidance for designing multimodal learning analytics systems to better align with and support their pedagogical practices in movement-based learning.

Keywords: Movement-based learning · Embodied learning · Multimodal learning analytics · Teacher cognition · Preservice teachers.

1 Introduction

Movement-based and embodied learning strategies are increasingly valued for their ability to integrate physical activity with cognitive tasks, helping students maintain engagement and internalize abstract concepts [15, 7]. However, such activities create dynamic environments with complex student behaviors (including body movements, facial expressions, hand gestures, and speech) that can easily exceed a teacher's capacity to monitor and interpret in real time [5, 18].

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While Multimodal Learning Analytics (MMLA) systems offer a promising solution to capture and synthesize these signals into actionable insights, their adoption in real classrooms remains limited. A key barrier to adoption is the lack of teacher perspectives in system design [12]. For MMLA systems to serve as effective, supportive agents to teachers, they must align with teachers' pedagogy, including their noticing and sense-making practices. To address this gap, we explore what future teachers look for during movement-based learning (using a 'design for the future' perspective) and identify how multimodal data can best support their monitoring and instruction. More specifically, in this study, we address the following research questions:

RQ1 How do future PreK–6 teachers interpret student behaviors (body movements, facial expressions, hand gestures, and speech) as evidence of understanding or lack of understanding during movement-based learning?
RQ2 What concerns do future PreK–6 teachers have during movement-based learning, and how do they anticipate responding to those concerns?
RQ3 What multimodal student data do future PreK–6 teachers perceive as most useful for monitoring and supporting students during movement-based learning?

2 Background

Movement-based learning broadly refers to pedagogical strategies that integrate physical activity into cognitive tasks, often to improve engagement or provide concrete experiences for abstract concepts [15]. **Embodied cognition** extends this view by emphasizing the role of the body in learning, arguing that cognition arises not just from the mind, but also through our bodies' physical and social interactions with the world [7]. In math education, movement-based and embodied learning help students internalize abstract concepts by directly linking physical interaction with math reasoning [2, 9, 15, 4, 3].

The role of physical interaction can take several forms along the physicality dimension of embodied activities, including direct embodied, enactment, manipulation, surrogate, and augmented [7]. This study focuses on the first three dimensions: activities where the body physically *represents* math concepts, *enacts* knowledge through movement, or *manipulates* objects to explore math ideas[1].

By making cognitive processes visible through physical interaction, movement-based learning creates a highly dynamic environment with complex student behavioral signals (*e.g.*, body movements, facial expressions, hand gestures, and speech) that can challenge teachers. While Cognitive Load Theory is traditionally used to explain how limited working memory restricts *student* information processing [20], these same constraints likely apply to *educators*. When the complexity of these signals exceeds a teacher's ability to monitor and interpret them,

[1] While three activities in this study incorporate technology, they do not meet the criteria for augmented embodiment defined by [7] and are not categorized as such.

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it can raise concerns about self-efficacy as well as the time, support, and resources needed to implement and manage the activity effectively [5, 18].

To help teachers monitor and interpret student behavioral signals in real time, **MMLA systems** have emerged as promising tools for collecting, analyzing, and visualizing data from multiple student activity channels [17, 19]. Often delivered through teacher dashboards [6, 12, 19, 10], these systems aim to reduce the teacher's cognitive burden by synthesizing students' raw physical, social, and technological interactions into actionable pedagogical insights. In this way, MMLA systems serve as a 'pedagogical companion' to support teachers' ability to notice and respond to students in dynamic learning environments [19], helping teachers to identify struggling students, develop personalized learning experiences, and optimize pedagogical practices [16].

Depending on the technology and modalities involved, MMLA systems can leverage a wide range of data collection and processing techniques to provide a comprehensive view of the embodied learning process [16]. For example, in a movement-based math activity, a system might track students' gaze using eye-tracking glasses, physiological data using a wristband, and facial expressions and body pose using a web camera. These data are aggregated to visualize metrics such as student cognitive load, perceived difficulty, and engagement [6].

Despite the promise of movement-based learning and MMLA, their use has largely been confined to research settings, with limited adoption in real classrooms. One contributing factor is the lack of research on teachers' perspectives regarding these systems [12]. If MMLA systems are to function as true 'pedagogical companions,' they must support, rather than override, teachers' instructional practices, aligning with their noticing and sense-making processes. To address this gap, the present study examines how preservice teachers (*i.e.*, the next generation of teachers) interpret student behavioral signals as evidence of understanding or lack of understanding, as well as their concerns and anticipated responses during movement-based learning activities.

Limited classroom adoption is also influenced by practical concerns around data collection [12, 16]. In this study, we investigate which types of multimodal student data future teachers are comfortable collecting, and which data they perceive as most useful for monitoring and supporting students, thereby informing what signals should be prioritized in MMLA systems for future teachers.

Finally, prior studies on movement-based and embodied learning often focus on a single activity, which limits understanding of how teachers interpret behaviors and respond across diverse tasks. In this study, we investigate teachers' perspectives across six activities that vary in math complexity and embodiment to capture a broader range of contexts.

3 Methodology

3.1 Participants and Activities

Participants include 25 preservice teachers enrolled in an upper-level undergraduate course in the College of Education at a large public university in the United

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States. The course focused on methods, materials, and curricular approaches for teaching math in grades PreK–6. Prior to the study, all participants had in-classroom teaching experience as teaching assistants and/or lead teachers, and 19 additionally reported out-of-classroom teaching experience (Table 1).

Table 1. Participants' teaching experiences (n = 25).

Teaching Experience	**n**	**(%)**
In-classroom	**25**	**(100)**
PreK–Kindergarten	22	(88)
Grades 1–2	17	(68)
Grades 3–4	9	(36)
Grades 5–6	4	(16)
Out-of-classroom	**19**	**(76)**
Afterschool programs	12	(48)
Summer camps	12	(48)
Tutoring	4	(16)
Educational volunteering (*e.g.*, museums)	2	(8)
Other	5	(20)

During the first week of the course, participants were introduced to movement-based learning and asked about their experiences with such activities. Eleven participants reported some experience, nine reported none, and five were unsure. Over the following 14 weeks, participants engaged in six biweekly movement-based math activities that varied in complexity and embodiment (Table 2). Each activity lasted approximately 20–30 minutes and took place during regular class sessions. While the activities and post-activity questionnaires were required for the course, participants had the option to consent to sharing their responses with researchers for each individual activity. To ensure voluntary participation, the course instructor was blinded to students' consent choices, and no incentives or penalties were associated with consenting or declining. The study was reviewed and approved as exempt by the University's Institutional Review Board.

The post-activity questionnaire included open-ended and Likert scale items, and took approximately 15–20 minutes to complete. The first question asked participants to describe the activity in their own words to verify their understanding. For the remaining questions, participants were told *Imagine you are running Activity in your classroom. Please answer the following questions specifically about your experiences with Activity and how you would lead it.* See Table 3 for questionnaire items.

Additionally, we wanted to gauge participants' comfort regarding student data collection. In the first questionnaire, we additionally asked participants *How comfortable would you be with <log data / eye-tracking or gaze detection / physiological signals / audio recordings / video recordings / motion or position tracking> being collected of your students in your classroom?*

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Table 2. Summary of the six movement-based learning activities, highlighting primary embodiment (*italicized*) and math concepts (**bold**) identified by the authors. All activities involve peer collaboration. Asterisks (*) indicate the use of mobile devices to provide problems and feedback through the embodied learning system Wearable-Learning (WL) [2, 4, 14]. The number of participants (n) varies across activities due to absences, resulting in a total of 119 questionnaire completions (N).

Activity	**Description**	**n**
Hopscotch	Students hop on numerical tiles along a grid, coordinating single- and double-foot hops to *enact* **counting** and **numerical order**.	25
Integer Hopscotch*	Students walk along a number line, guided by problems provided by the WL app, using their bodies to *represent numbers* on the number line and *enact* **addition and subtraction with positive and negative integers**.	22
Tangrams Race*	Students run relay-style to collect Tangram shapes based on math clues shown on the WL app, first *representing shapes* with their hands to depict the correct piece, and then *manipulating* all collected pieces to assemble the final Tangram puzzle, practicing **geometry** and **spatial reasoning**.	14
Estimate It!*	Students search the classroom for 3D shapes that match math clues shown on the WL app, *manipulating* the shapes and using an unmarked 12-inch dowel to **estimate geometric measurements**.	19
Ghosts Number Line	Students arrange ghost cutouts with fractions, decimals, and percents on a clothesline that represents a number line from 0 to 1, *manipulating* them to ensure correct placement and ordering, practicing **rational numbers** and **equivalences between fractions, decimals, and percents**.	16
Building Volumes: Icosahedron	Students construct a 3D icosahedron from nets of triangles using paper, magnets, or dowels, *manipulating* the materials to connect edges and build the 3D shape, exploring its structure and practicing **geometry with polyhedra** and **spatial reasoning**.	23

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Table 3. Post-activity questionnaire items asked at the end of each activity. Brackets (<>) denote variable elements systematically permuted for the final questionnaire.

Item	Type	Purpose
What <*body movements* / *facial expressions* / *hand gestures* / *speech*> might suggest to you that a student <*understands* / *does not understand*> the math concepts during *Activity*?	Open-ended	*RQ1*
What are your concerns during *Activity*?	Open-ended	*RQ2*
What student behaviors or situations would you want to be alerted to during *Activity*?	Open-ended	*RQ2*
Where do you think students might struggle most during *Activity*, and how would you respond as the teacher?	Open-ended	*RQ2*
How useful would <*body movements* / *facial expressions* / *hand gestures* / *speech*> be in helping you <*monitor* / *support*> students during *Activity*?	Likert scale	*RQ3*
Imagine a tool designed to support you during *Activity*. How useful would displaying <*log data* / *gaze or attention direction* / *physiological signals* / *voice or speech patterns* / *collaboration* / *postures and gestures* / *facial expressions* / *motion and position*> be for supporting your pedagogical practices.	Likert scale	*RQ3*

3.2 Data Analysis

To analyze responses, we used a combination of qualitative and quantitative methods. To address *RQ1*, participants responded to questions varying along two dimensions: (1) **behavioral signals** (*body movements*, *facial expressions*, *hand gestures*, *speech*) and (2) **understanding** (*understands*, *does not understand*). This yielded eight behavior–understanding questions. We analyzed responses to these questions using inductive coding supported by affinity diagramming.

For the first three activities, we constructed one affinity diagram for each of the eight behavior–understanding questions, resulting in 24 affinity diagrams. We then synthesized themes identified across these diagrams into a single coding dictionary, which we then applied systematically across all six activities, including the first three activities again, for each behavior–understanding question. We allowed the dictionary to remain flexible to accommodate new categories if previously unobserved signals emerged in later activities. This inductive approach ensured that our coding reflected participants' actual words and experiences while maintaining consistency and supporting comparability across activities.

To address *RQ2*, participants responded to three open-ended questions per activity. Unlike the analysis conducted for *RQ1*, we aggregated responses for each of these questions across all activities to construct three high-level affinity diagrams. We adopted this high level of abstraction after finding little variability in responses across activities for *RQ1*.

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Finally, to address *RQ3*, participants responded to a series of five-point Likert scale questions (with an additional unsure option) that we analyzed using descriptive statistics.

4 Results

4.1 Interpreting Student Behaviors to Gauge Understanding

Based on responses across activities, we identified eight distinct dimensions of student behavioral signals that future teachers use to gauge understanding: *affect*, *correctness*, *focus and engagement*, *intention*, *peer interaction and collaboration*, *rhythm and confidence*, *support use*, and *technology use* (Figure 1). These dimensions were synthesized directly from the open-ended responses. For example, while *focus* and *engagement* are theoretically distinct, participants consistently discussed them in parallel and thus are mapped as a single dimension.

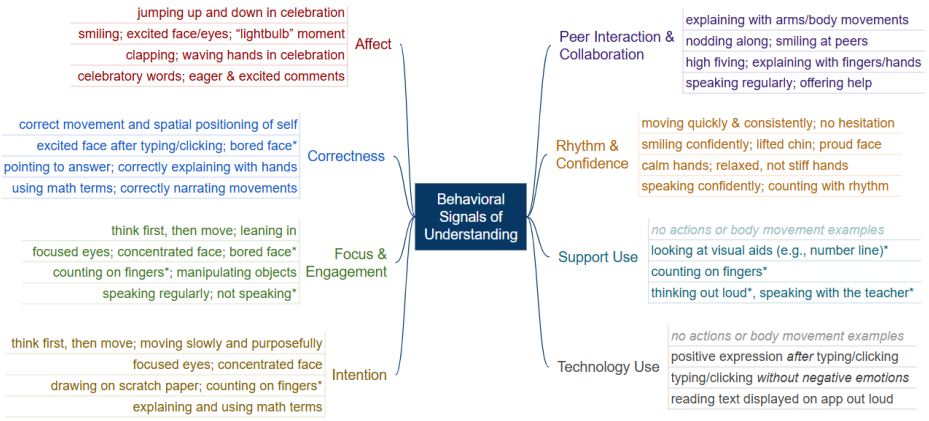


Fig. 1. Word tree of eight dimensions of signals for gauging student understanding. Within each dimension, representative responses are ordered by signal (top to bottom: body, face, hand, speech). Asterisks (*) denote ambiguous behaviors cited as evidence for both understanding and lack of understanding; others follow an inverse relationship.

For most signals, future teachers interpret understanding and lack of understanding as an inverse relationship. For example, within the *affect* dimension, "jumping up and down" and "smiling" indicate understanding, while "dropping arms and being still" and "frowning" indicate lack of understanding. Similarly, in *rhythm and confidence*, "moving quickly and consistently" indicates understanding, while "moving slowly and hesitantly" indicates lack of understanding.

However, there are some signals that could be evidence of understanding or lack of understanding, depending on pedagogical perspectives. We refer to these signals as *ambiguous signals* (asterisked in Figure 1). For example, some

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participants indicate a "bored face" signals mastery, while others argue it signals disengagement or confusion. Similarly, "not speaking" could signal deep focus or fear of being wrong. And, "support use", like using fingers to count or visual aids, could indicate understanding, though some argue these aids signal confusion because students with deep understanding should not need such scaffolding.

4.2 Concerns and Anticipated Responses

Participants reported a range of concerns during movement-based learning (Figure 2). The most frequent concern across all activities is *understanding* (50.9%), followed by *affective states* (15.1%) and *engagement* (11.3%). While *safety* shares the same global percentage as *engagement*, the majority of *safety* concerns (61.1%) come from a single activity: Hopscotch. We hypothesize this is because Hopscotch targets young students, and future teachers may prioritize their safety more than that of older students who typically have more developed motor skills.

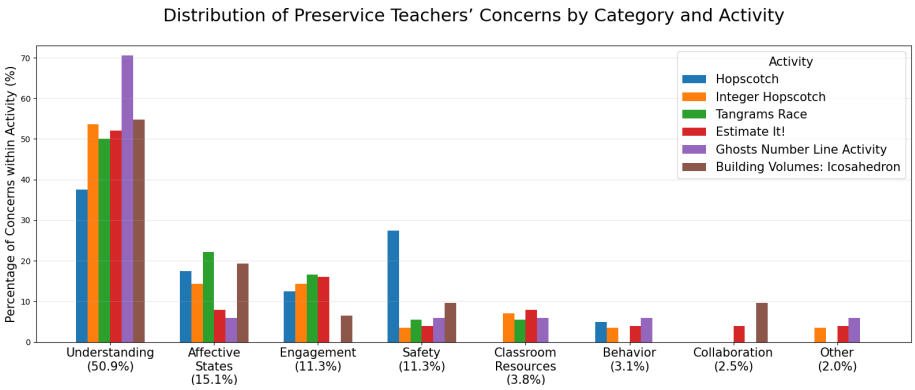


Fig. 2. Distribution of preservice teachers' concerns regarding movement-based learning activities across categories. Parentheses indicate global percentages.

Given the predominance of *understanding* concerns, we further analyzed this category and identified three main sub-categories: understanding of *math concepts* (53.7%), understanding of the *activity* (25.6%), and understanding of the *connection between movements and math concepts* (4.9%).

As shown in Table 4, future teachers anticipate aligning their responses to the cognitive, emotional, and motivational dimensions of student struggle. For *understanding* concerns (cognitive), they plan to provide active support through modeling and scaffolding. For *affective* concerns (emotional), they plan to primarily use interpersonal strategies, such as reassurance and encouragement, supplemented with active support as needed. For *engagement* concerns (motivational), they plan to continuously monitor students and adapt tasks responsively. Overall, these strategies illustrate an integrated approach to addressing their concerns across the cognitive, emotional, and motivational dimensions of student struggle.

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Table 4. Primary concerns and anticipated response strategies of preservice teachers during movement-based learning activities.

Concern	Anticipated Response Strategies
Understanding	
of math concepts	Modeling step-by-step with visuals; Clarifying/reinforcing instructions; Scaffolding with hints and guiding questions
of activity	Demonstrating activities or specific movements; Clarifying/reinforcing instructions; Simplifying tasks and movements
of movement–math connection	Explaining the purpose of movements while demonstrating
Affective States	Calming and reassuring students; Encouraging persistence; Providing hands-on support; Clarifying to reduce frustration
Engagement	Mixing up rules and difficulty; Monitoring/checking-in every few minutes; Providing alternatives for technical issues

4.3 Perceived Usefulness of Multimodal Student Data

We first analyzed the overall comfort of future teachers toward various multimodal data collection streams (Figure 3). Median and mode ratings for nearly all streams indicate participants are Somewhat to Very Comfortable, with *motion or position tracking* being the only exception (median = Neutral). Overall, this pattern suggests an emerging professional readiness among future teachers to use systems that collect multimodal classroom data.

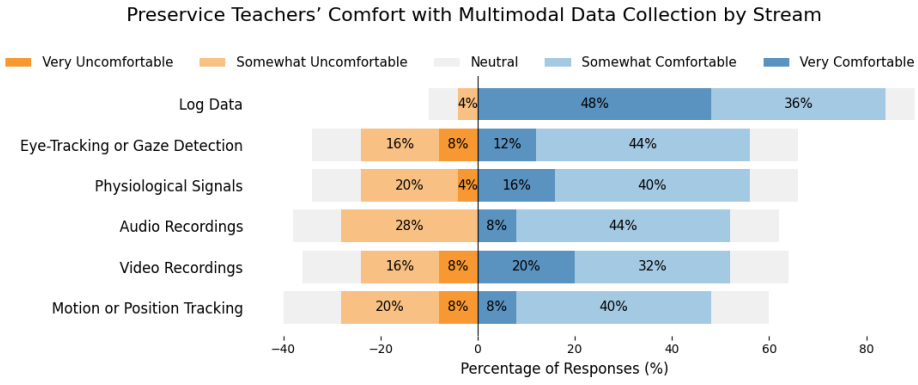


Fig. 3. Preservice teachers' self-reported comfort with various multimodal data collection streams. Bars diverge from the center, with the strongest discomfort or comfort toward the middle and neutral responses split equally on the outer edges.

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Building on this baseline of professional readiness, we next evaluated how preservice teachers perceive the usefulness of the signals derived from these streams. We first examined **general** signals to compare their usefulness for specific pedagogical purposes (*monitoring* vs *supporting* students; Figure 4). We then analyzed signals at a more **granular** level to identify which individual data types are perceived as most useful overall (Figure 5).

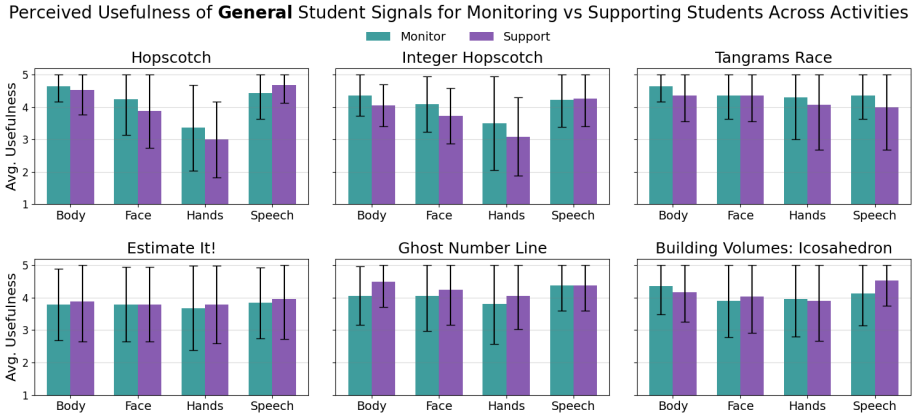


Fig. 4. Preservice teachers' perceived usefulness of **general** student signals across movement-based activities (1 = Not at All, 5 = Extremely) by pedagogical purpose (*monitoring* vs *supporting*). Error bars represent standard deviations capped at 5.

For monitoring students, a complementary relationship exists between body and speech signals. In the **general** analysis, *body movements* and *speech* are consistently perceived as the two most useful signals, alternating as the primary depending on the activity. Specifically, in four of the six activities (Hopscotch, Integer Hopscotch, Tangrams Race, and Building Volumes), *body movements* is first-rated, with *speech* second. In the remaining two activities, *speech* is first-rated, with *body movements* second. This pattern is further supported in the **granular** analysis, where *postures and gestures* and *motion and position* are first-rated for the same four activities as *body movements*, and *collaboration* is first-rated for the same two activities as *speech*.

For supporting students, a similar complementary relationship between body and speech is observed; however, speech is rated as more useful than body in most activities (Hopscotch, Integer Hopscotch, Estimate It!, and Building Volumes). Tangrams Race is the only exception, where *body movements* and *facial expressions* are tied as first-rated, and *speech* is rated lowest. We hypothesize that in this relay-style game, students mainly used speech to cheer on peers until the final collaboration, making it less important than other signals.

Collectively, these findings demonstrate that body movements and speech function as a core 'multimodal duo' for monitoring and supporting students.

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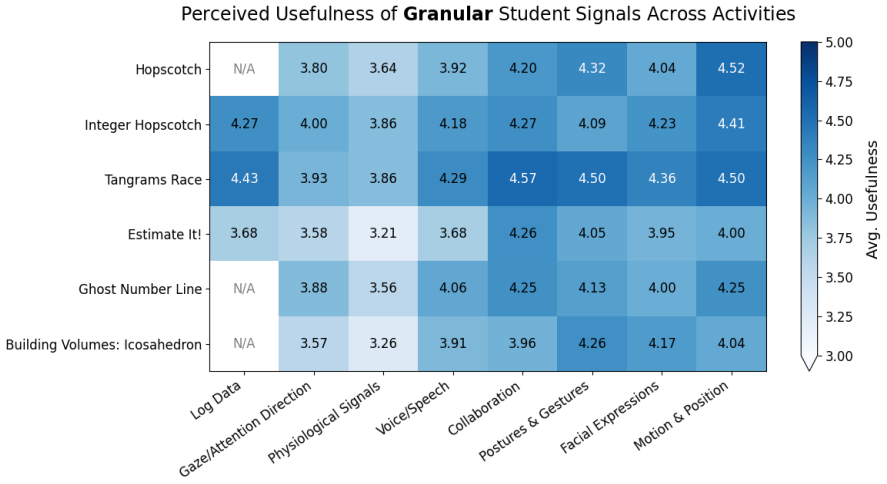


Fig. 5. Preservice teachers' perceived usefulness of **granular** student signals across movement-based activities (1 = Not at All, 5 = Extremely) for broad pedagogical purposes. Scale is truncated at 3.0 to better highlight relative differences.

While their relative priority shifts slightly according to the learning activity, their combined high utility establishes a clear design requirement for future MMLA systems aiming to support teachers during movement-based learning activities.

5 Discussion

In this study, we identified eight dimensions of behavioral signals that future teachers use to gauge student understanding during movement-based learning. Although our analysis was conducted at the activity level, we observe no significant differences in signals across activities, despite variation in math complexity (PreK to Grade 6 concepts) and embodiment (direct embodiment, enactment, manipulation). We find this result notable and hypothesize that these dimensions constitute a foundational *observational framework* that future teachers draw upon to interpret the embodied learning process, largely independent of specific activity characteristics. This finding aligns with prior work in MMLA design emphasizing the need to balance contextual specificity with generalizable components, suggesting that while learning scenarios differ, common behavioral patterns can support more broadly applicable designs [8]. Together, these results suggest MMLA systems may not need activity-specific tailoring, but could focus on robustly capturing core behavioral dimensions across activities.

While most behavioral signals show a clear inverse relationship between understanding and lack of understanding, ambiguous signals exist and pose a challenge for MMLA system design. Although prior work emphasizes teacher-in-the-loop approaches that center pedagogical perspectives in MMLA system design [12], our findings raise a complex design question that challenges how we can

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accomplish this: *how can MMLA systems accommodate conflicting pedagogical perspectives of the same student behavior?*

Research has begun to address MMLA system design considerations [8]; however, we found no work that explores how teachers can customize the interpretation of behavioral signals. For example, a teacher might wish to configure a dashboard such that boredom signals mastery for advanced students and disengagement for others. We hypothesize that supporting such pedagogically grounded customization could increase teacher trust and adoption by embedding professional judgment into the system. We encourage future work to explore MMLA system designs that support such adaptive, teacher-configurable sense-making.

In addition to trust, ethical and privacy considerations are additional factors that slow the adoption of MMLA systems by teachers [1, 12, 16]. Although our findings suggest future teachers are generally comfortable with various forms of student data collection, concerns related to how data are collected, processed, and managed remain salient. Our results should not be interpreted as an endorsement of unrestricted data collection, but rather as an indicator of professional readiness of future teachers to engage with data-informed tools such as MMLA systems. Importantly, there is a strong need for clear data governance practices within MMLA systems, including transparency around who has access to student data, how long data are stored, and how they may be reused. Additionally, issues of consent in classroom settings remain complex, as students may have limited ability to meaningfully opt out of multimodal data collection when such systems are embedded into required activities [13]. These challenges are further compounded by classroom power dynamics and the interpretive nature of behavioral data; as our findings show, ambiguous behavioral signals may be understood differently across teachers, raising the risk of misinterpretation or bias if such signals are treated as objective indicators. Future work must continue to prioritize ethics and privacy of student data before large-scale MMLA system adoption can occur [13, 8].

While issues of trust, ethics, and privacy are all major concerns identified in prior research, this study identifies an overwhelming concern centered on student understanding, accounting for around half of all responses (50.9%). This finding highlights that cognitive sense-making remains the primary focus of future teachers during movement-based learning. Furthermore, these understanding concerns are multifaceted: teachers attend to understanding of math concepts, understanding of the activity, and understanding of movement–math connection. This finding suggests that MMLA systems must be designed to capture and communicate multiple dimensions of understanding, rather than focusing solely on content mastery. Targeting the eight dimensions of behavioral signals identified earlier in this study could help capture these multiple facets.

Lastly, when monitoring and supporting students, future teachers primarily focus on body movements and speech. Across most activities, body movements are generally perceived as slightly more useful for monitoring than for supporting. Conversely, speech is generally perceived as slightly more useful for supporting than for monitoring. However, given that neither observation is significant, this

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further highlights that common behavioral patterns can support more broadly applicable designs [8]. However, there is a dip in perceived usefulness of hand gestures for Hopscotch and Integer Hopscotch as opposed to the other four activities. We hypothesize that this dip is the result of a lack of manipulatives in these activities. However, future teachers still notably find hand gestures somewhat useful for monitoring and supporting students even without manipulatives.

6 Conclusion

This study identifies eight dimensions of behavioral signals that future teachers use to interpret student understanding and lack of understanding in movement-based learning. While many signals show a clear inverse relationship, some are ambiguous, which could present challenges for MMLA system design in accommodating conflicting pedagogical perspectives. Despite variation in math complexity and embodiment across six activities, the consistency of these dimensions suggests a stable observational framework that could inform the design of generalizable MMLA systems.

This study has several limitations. Our findings are based on preservice teachers' reflections rather than real classroom observations, which may not fully capture student behaviors. While focusing on preservice teachers aligns with a 'design for the future' perspective, their limited experience may reduce generalizability, though all had some classroom experience and most had additional out-of-classroom experience. Additionally, we focused on PreK–6 grades, six specific activities, and specific math content, and thus our findings may not extend to other grades, activities, or subjects. Finally, we focused on core behavioral signals, omitting other student cues or contextual factors that may influence teacher interpretations.

There are two primary future directions that emerged from this work for MMLA research in general: (1) explore MMLA system designs that support adaptive, teacher-configurable sense-making, and (2) now that we see a professional readiness emerging, continue to explore and prioritize ethics and privacy of student data. Additionally, we plan to use the findings from this study to guide the evolution of an existing teacher dashboard prototype [11] to better support embodied learning and MMLA.

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